



Graph Matrix Representation

Graph Theory – Week 5

Ilham Gurat Adillion
ilhamgurata@its.ac.id



Ask yourself first:

Are you comfortable already with these concepts?

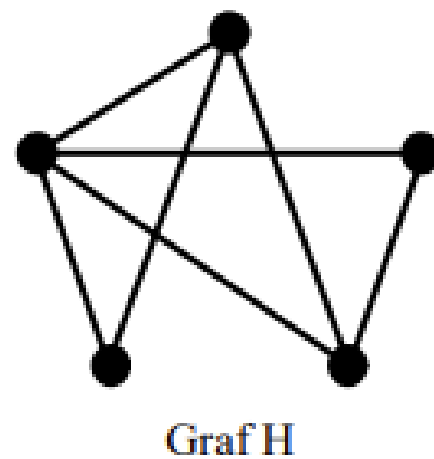
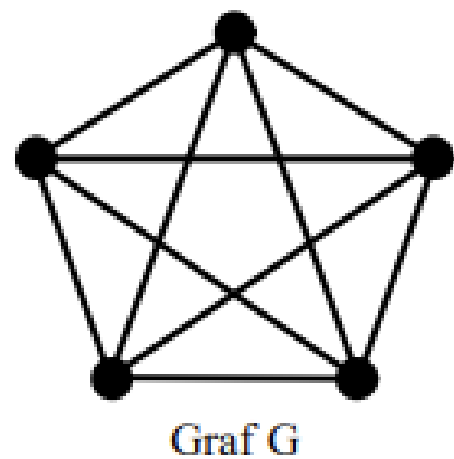
1. Graph glossary
2. Degree of a vertex

If anything still shaky, **ask now!**

Please also open week 1 glossary slide for keywords reference

Motivation

Graph is understandable in a visual form

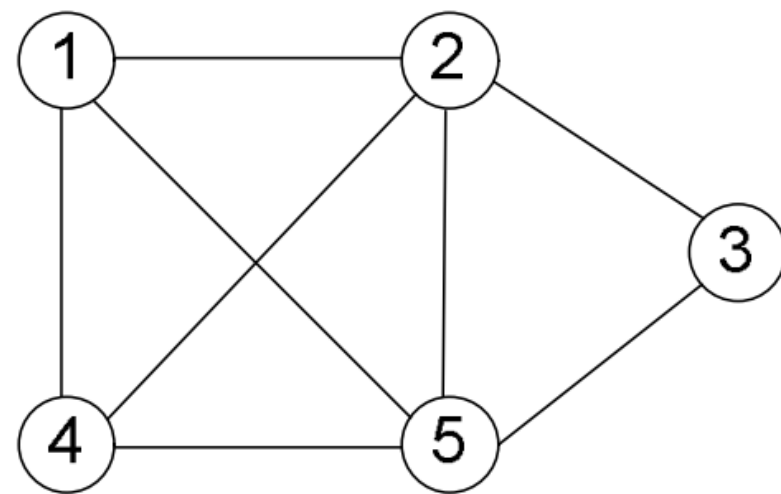


However, visual form isn't straightforward for computation/coding
The, how can we represent graphs for computation/coding problem ?

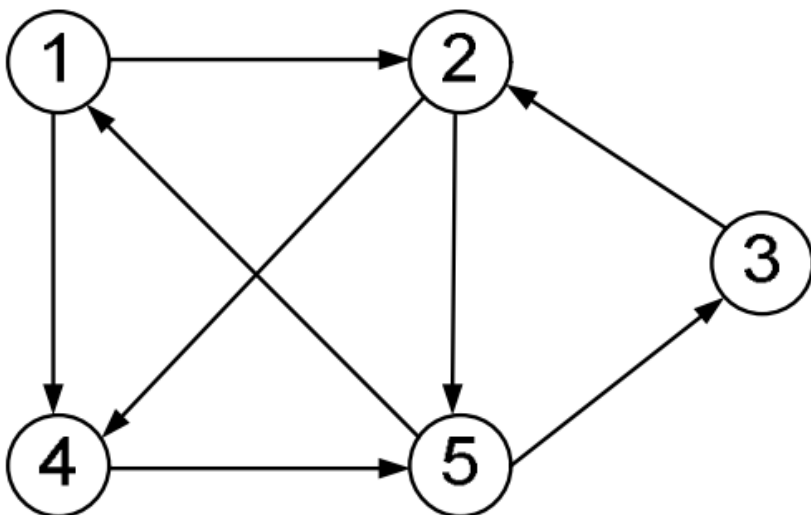


Adjacency Matrix

Give value 1 if there are any edge between two nodes. 0 otherwise.

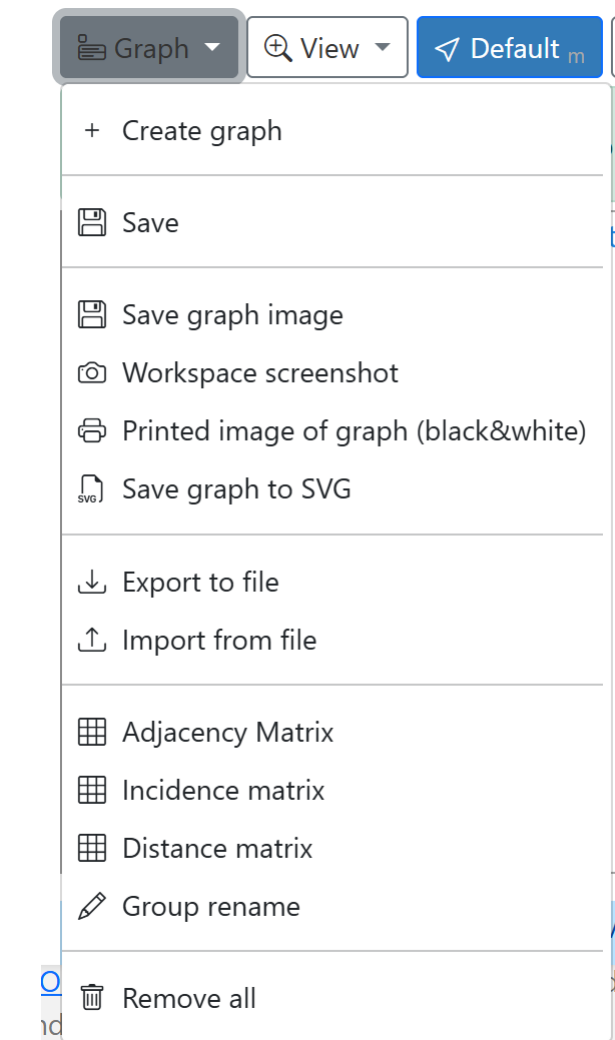


$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$



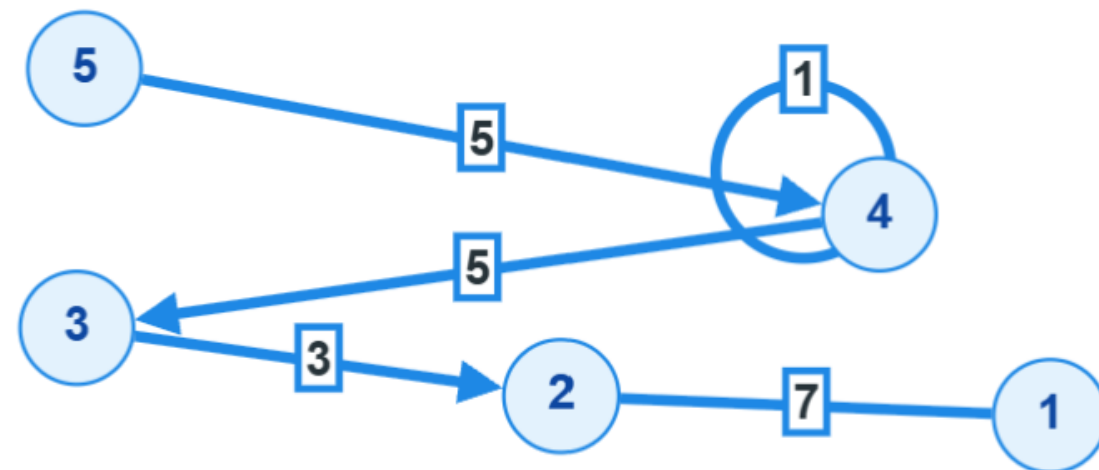
$$B = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Simulate @ graphonline.top



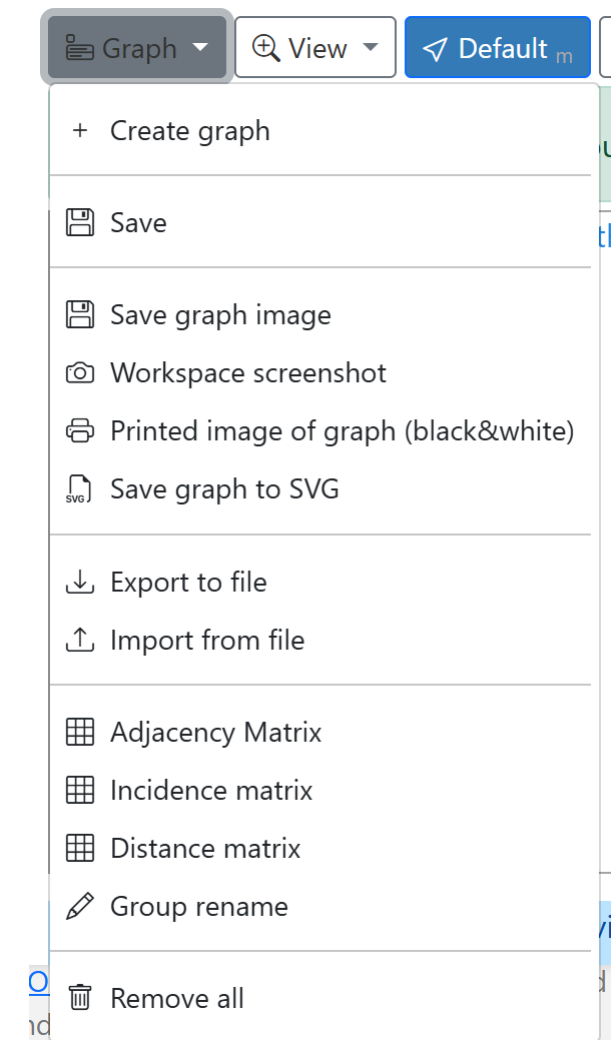
Weighted Adjacency Matrix

Give **weight value** if there are any edge between two nodes. **0** otherwise.



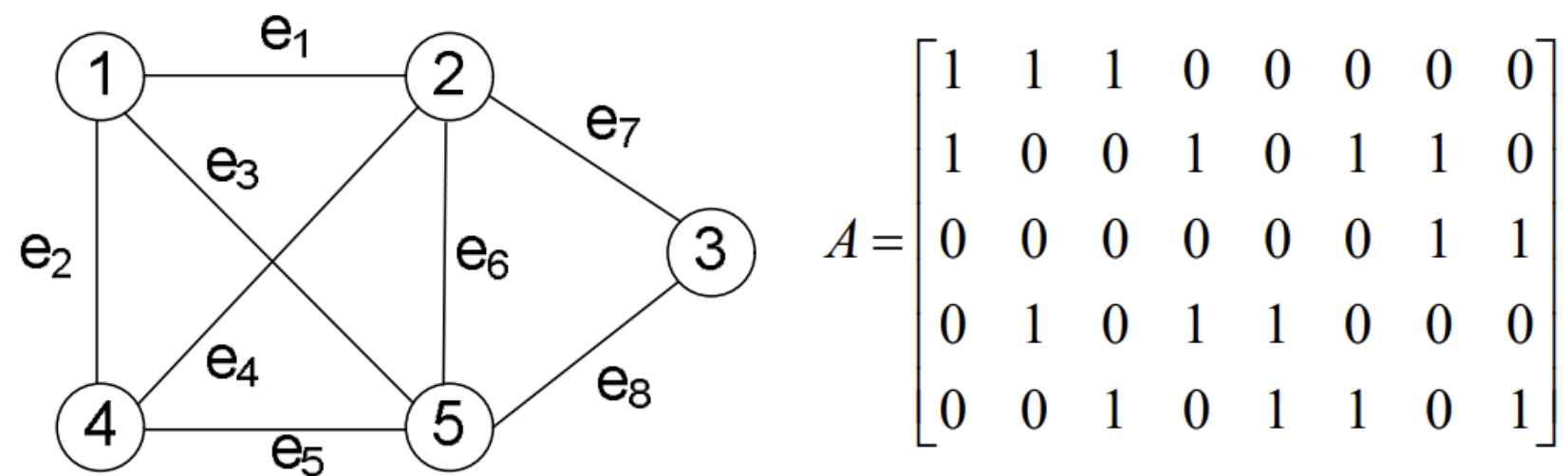
→	1	2	3	4	5
1	0, 7, 0, 0, 0,				
2	7, 0, 0, 0, 0,				
3	0, 3, 0, 0, 0,				
4	0, 0, 5, 1, 0,				
5	0, 0, 0, 5, 0,				

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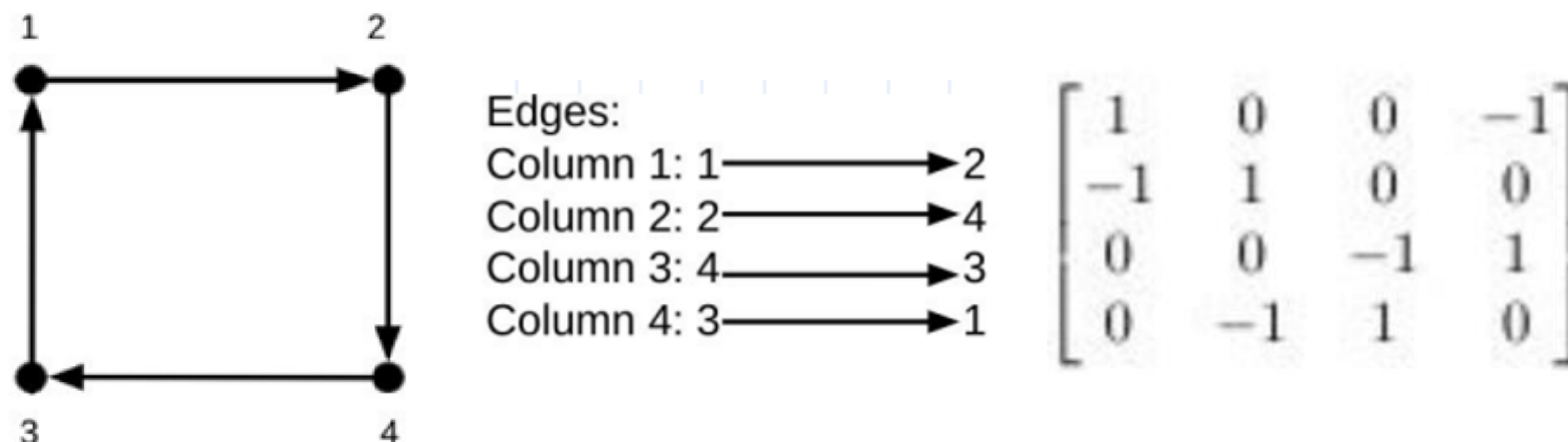


Incidence Matrix

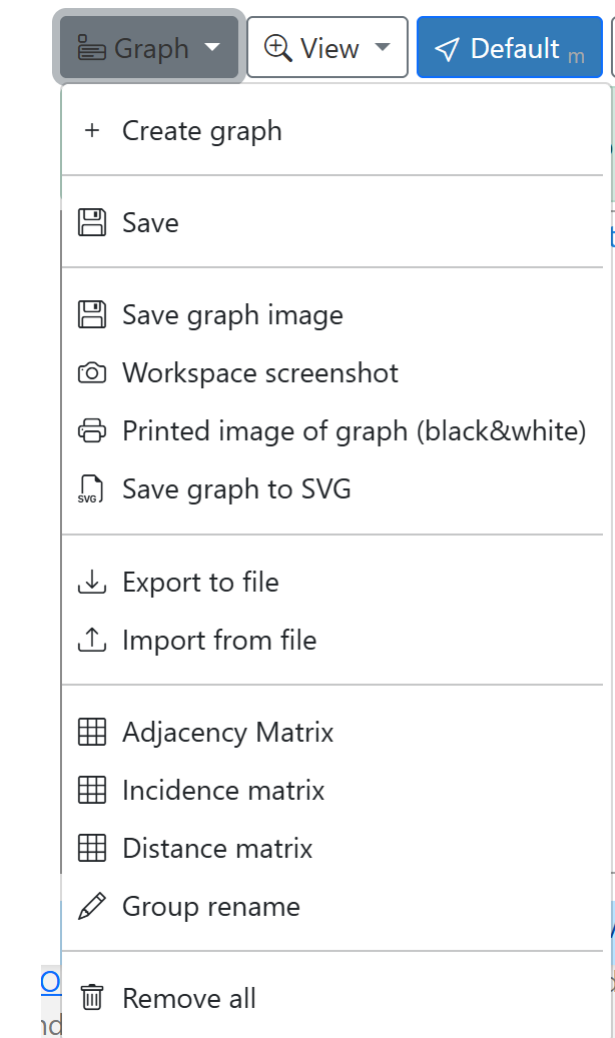
Give value 1 if a node has incidence with an edge. 0 otherwise.



For Digraph,
Give value 1 if an edge is “leaving” a vertex.
-1 if an edge is “arriving” at a vertex. 0 otherwise.



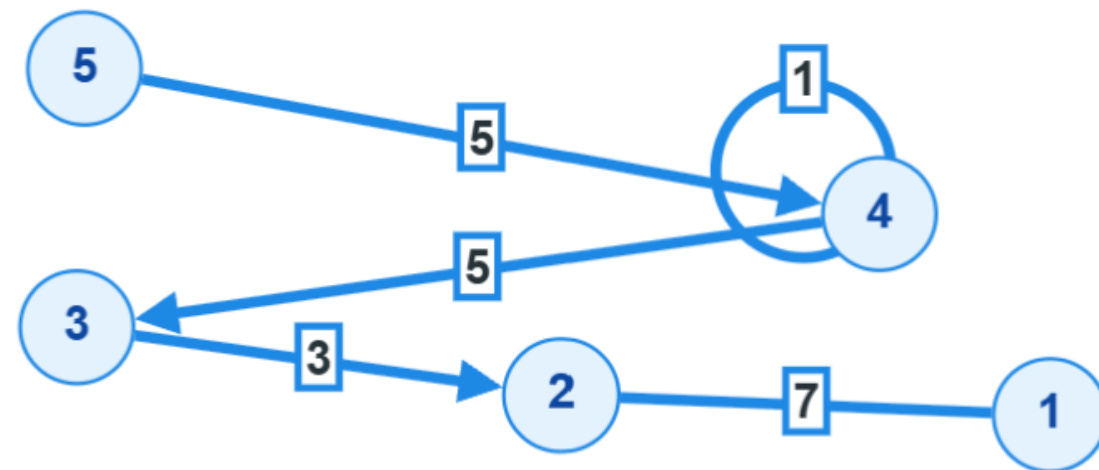
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Weighted Incidence Matrix

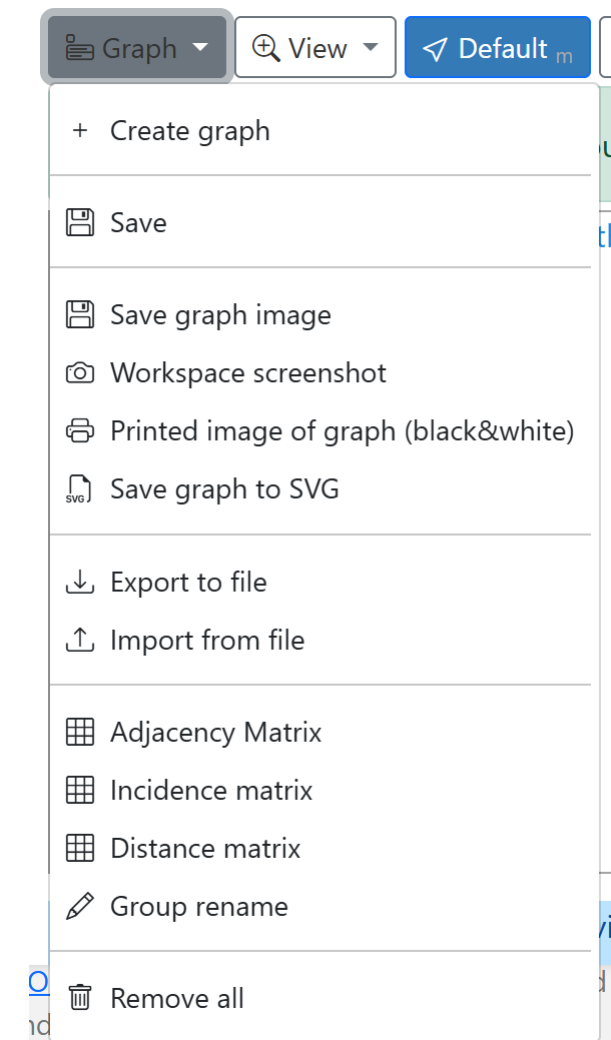
Give **weight value** if a node has incidence with an edge. 0 otherwise.

For digraph, weight is positive if “leaving” a vertex, and vice versa



1	7, 0, 0, 0, 0
2	7, -3, 0, 0, 0
3	0, 3, 0, -5, 0
4	0, 0, -5, 5, 1
5	0, 0, 5, 0, 0

Simulate @ graphonline.top



Fundamental Cycle Matrix

A fundamental cycle in a graph is formed when you take a spanning tree and add one more edge that's *not already* part of the tree

Motivation: Think of it as an independent alternative-route representation. Each non-tree edge reveals an additional route between two vertices through the spanning tree.

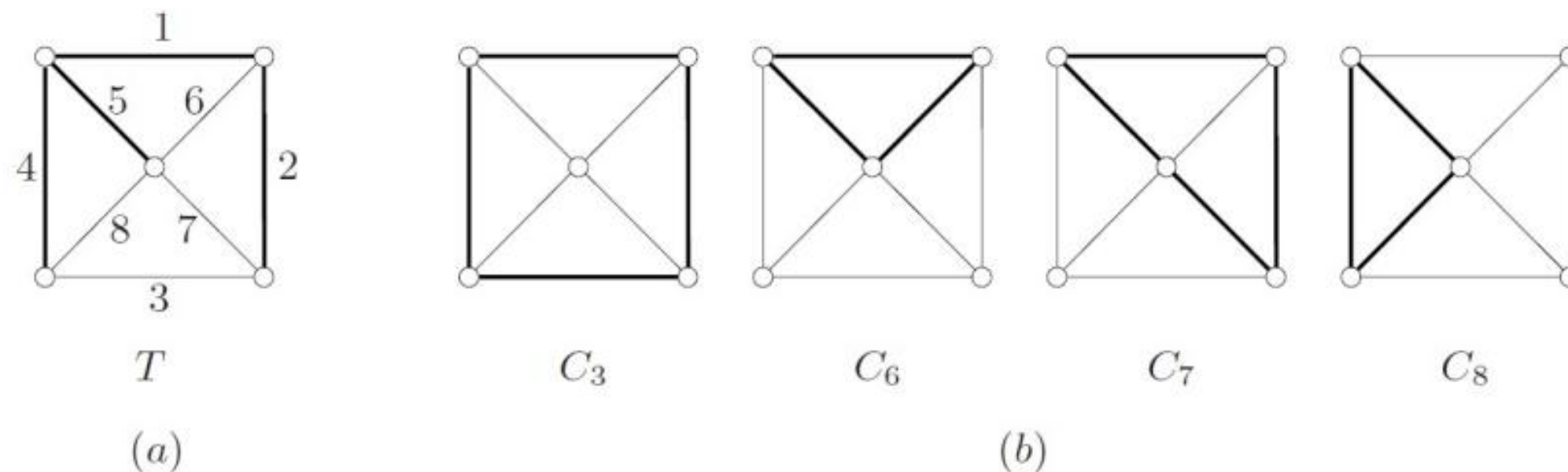
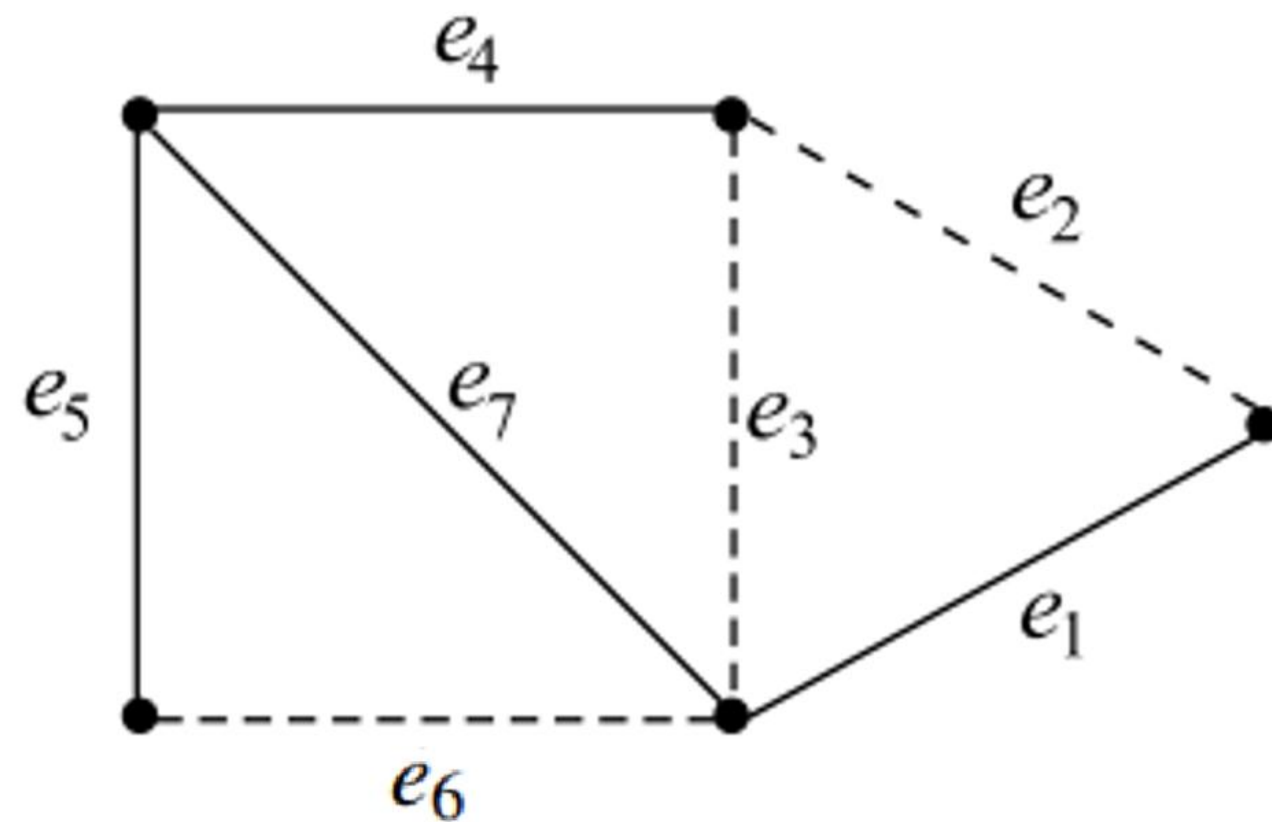


Figure 4.4. A spanning tree T and its four fundamental cycles.

Fundamental Cycle Matrix

Give value 1 if an edge is part of a fundamental cycle. 0 otherwise.



$$Z_1 = \{e_1, e_2, e_4, e_7\}$$

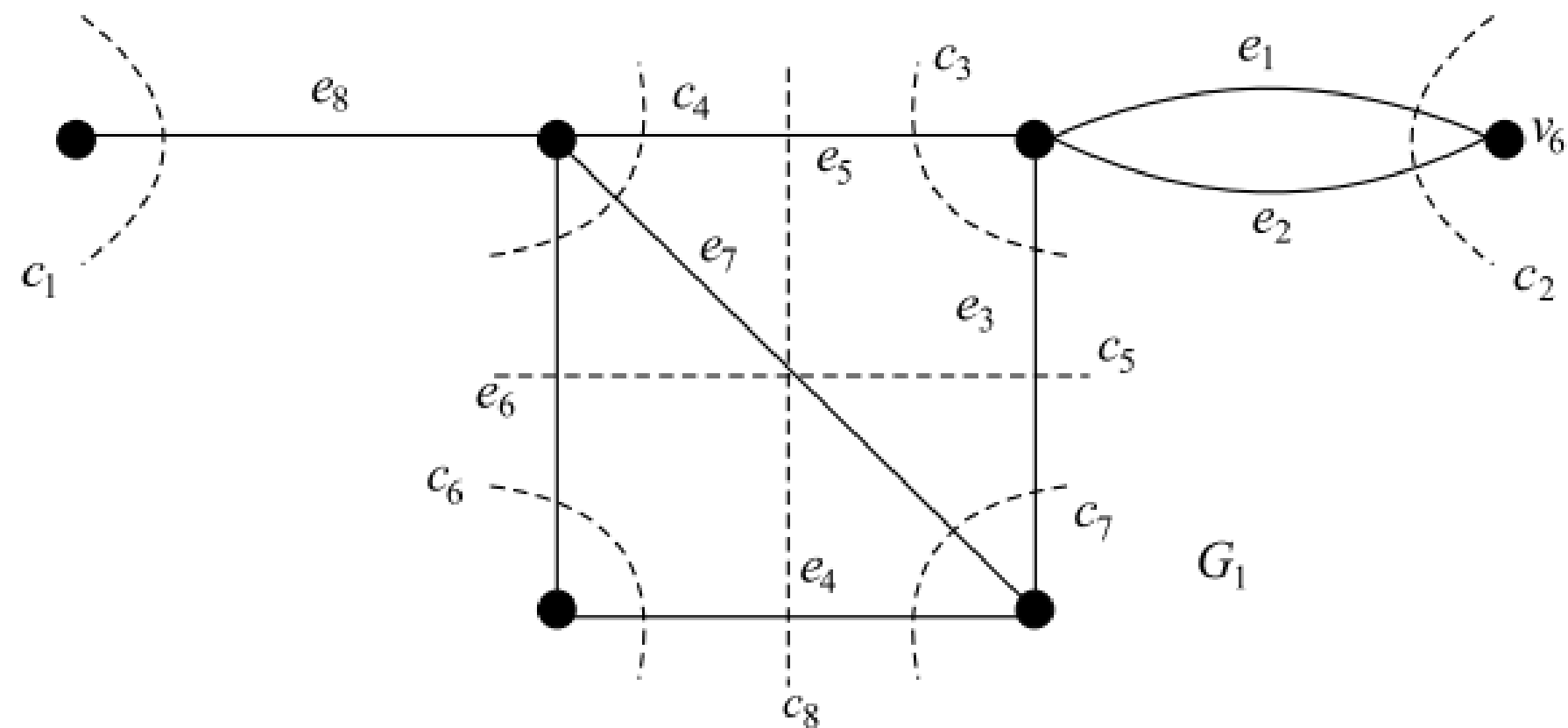
$$Z_2 = \{e_3, e_4, e_7\}$$

$$Z_3 = \{e_5, e_6, e_7\}$$

$$\begin{matrix} Z_1 \\ Z_2 \\ Z_3 \end{matrix} \begin{bmatrix} & e_2 & e_3 & e_6 & & e_1 & e_4 & e_5 & e_7 \\ 1 & 0 & 0 & \vdots & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & \vdots & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & \vdots & 0 & 0 & 1 & 1 \end{bmatrix}.$$

Cut-Edge/Cut-Set Matrix

Give value 1 if an edge is corresponding with a cut-edge/cut-set. 0 otherwise.

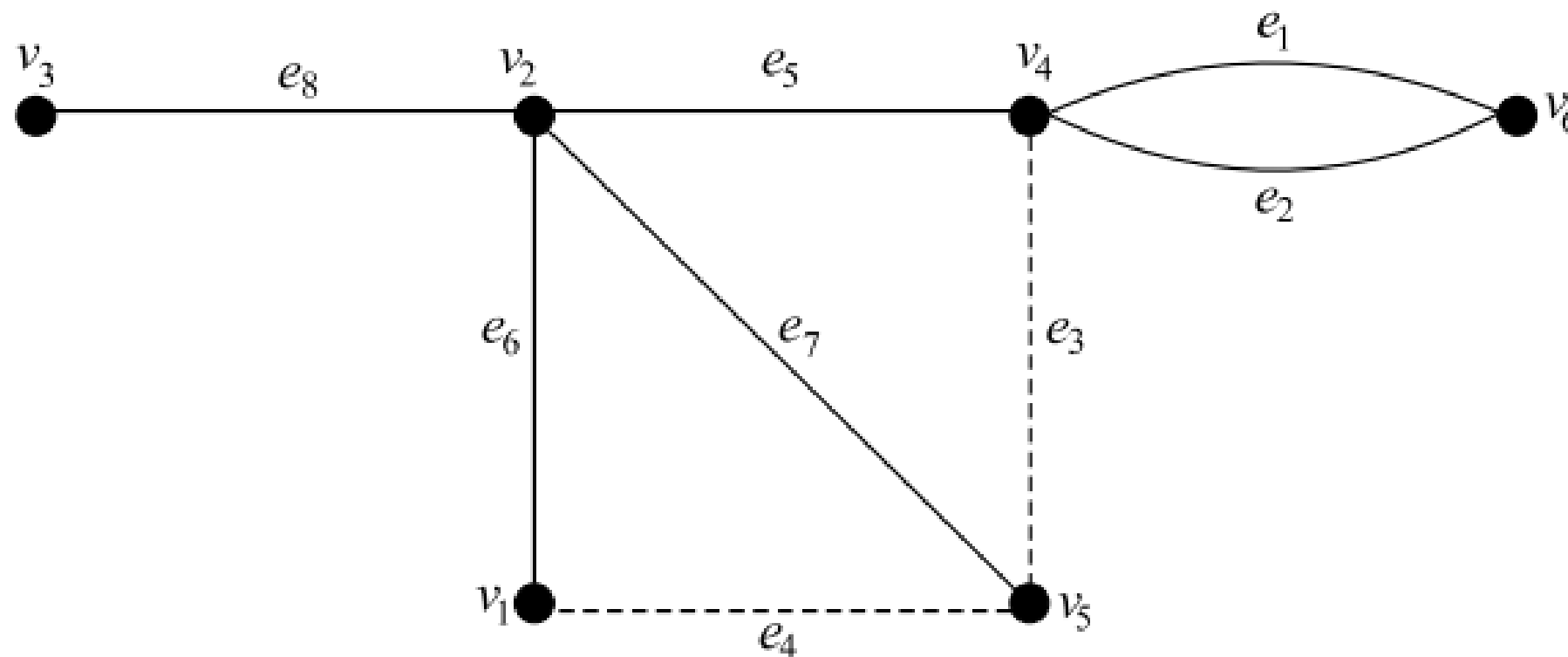


$$C(G_1) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{matrix} \\ \begin{matrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \\ c_8 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

The cut-sets are $c_1 = \{e_8\}$, $c_2 = \{e_1, e_2\}$, $c_3 = \{e_3, e_5\}$, $c_4 = \{e_5, e_6, e_7\}$, $c_5 = \{e_3, e_6, e_7\}$, $c_6 = \{e_4, e_6\}$, $c_7 = \{e_3, e_4, e_7\}$ and $c_8 = \{e_4, e_5, e_7\}$.

Path Matrix

Give value 1 if an edge is traversed by a path. 0 otherwise.



$$P(v_3, v_4) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{matrix} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

The different paths between the vertices v_3 and v_4 are

$$p_1 = \{e_8, e_5\}, p_2 = \{e_8, e_7, e_3\} \text{ and } p_3 = \{e_8, e_6, e_4, e_3\}.$$

Exercise

Your team is building an AI service with several components:

- | | |
|----------------------|---------------|
| A — Image Upload | D — Tracking |
| B — Preprocessing | E — Database |
| C — Object Detection | F — Dashboard |

A component can communicate directly with another component if connection exists.

The current architecture is: A–B, B–C, B–E, C–D, C–E, D–E, D–F, E–F

1. Draw the graph to represent it visually
2. Represent the architecture as an adjacency matrix
3. Analyse, Which components does E (Database) communicate with directly?
4. Which components have the highest number of direct connections?
5. If communication between C and E is disabled, update the matrix and determine which components are affected directly.

Exercise

The network of data centers is represented as an undirected graph G with the following structure:

Vertices: $V = \{1, 2, 3, 4, 5\}$

where each vertex represents a data center.

Edges: $E = \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 5), (4, 5)\}$

where each edge represents a communication link between data centers.

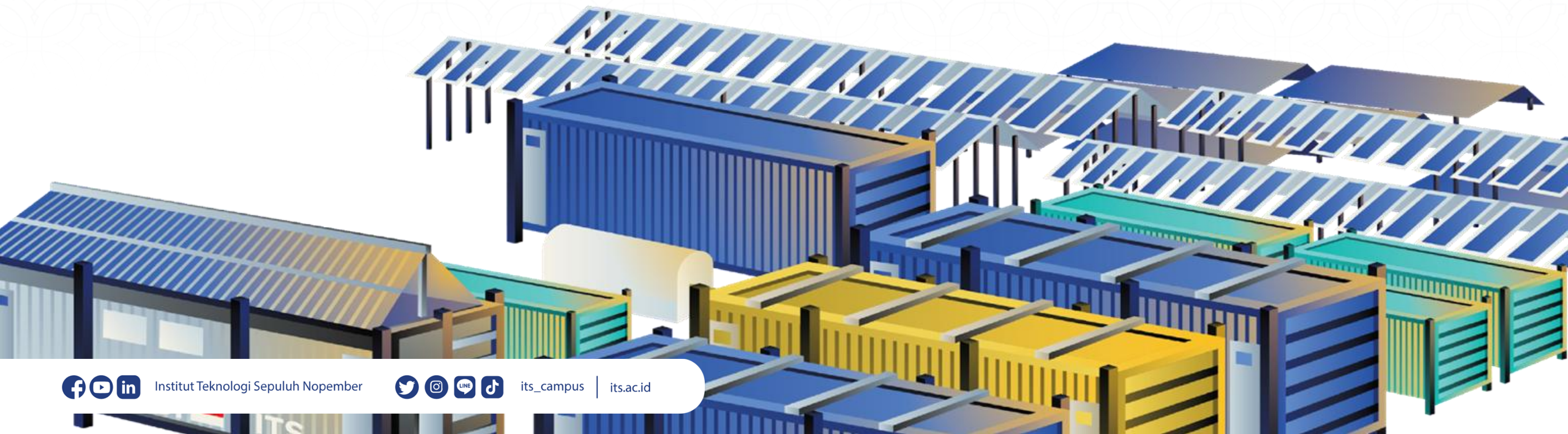
1. Construct the cut edge matrix C of the graph.
2. Which communication links are critical because their failure would disconnect a data center from the rest of the network?
3. For each critical communication link identified in 2), propose an additional communication link that will prevent that data center from becoming disconnected if the critical link fails. Explain your choice using the graph.

Homework

All homework in this course is a group homework (4 people)

Please submit your work on time or face credit reduction as a penalty.

Any use of AI is to be disclosed honestly by supplementing prompt history on submitted homework.



*Challenge yourself!
Don't blindly copy
AI answers!*

Group Homework

Graph Visualizer

Your task is to make following app:

1. User can input either adjacency or incidence matrix
2. The app visualize the graph
3. The app visualize and outputs the fundamental cycle matrix and cut-set matrix from inputted graph

Deliverables

GitHub repository, containing:

1. Code
2. README containing identity, prerequisite, how to run, sample input/output

Thank You

for your attention